

The importance of Seed Types for Tropical Cyclone Genesis in the Eastern and Central Pacific

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Background & Motivation

Tropical Cyclone Genesis (TCs) remains an open question in the research community; however, it is well-known that these factors are necessary for a storm to occur.

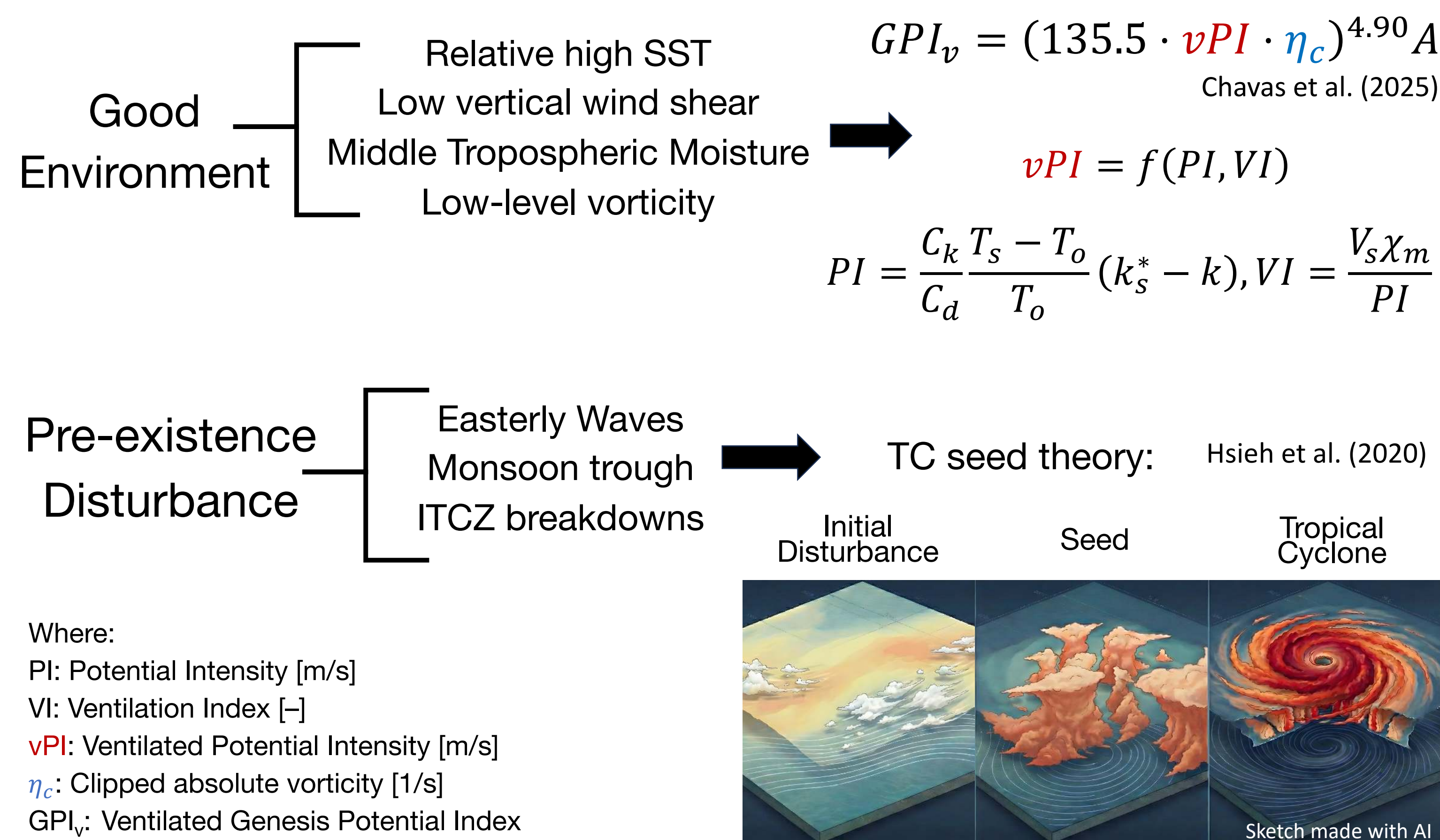
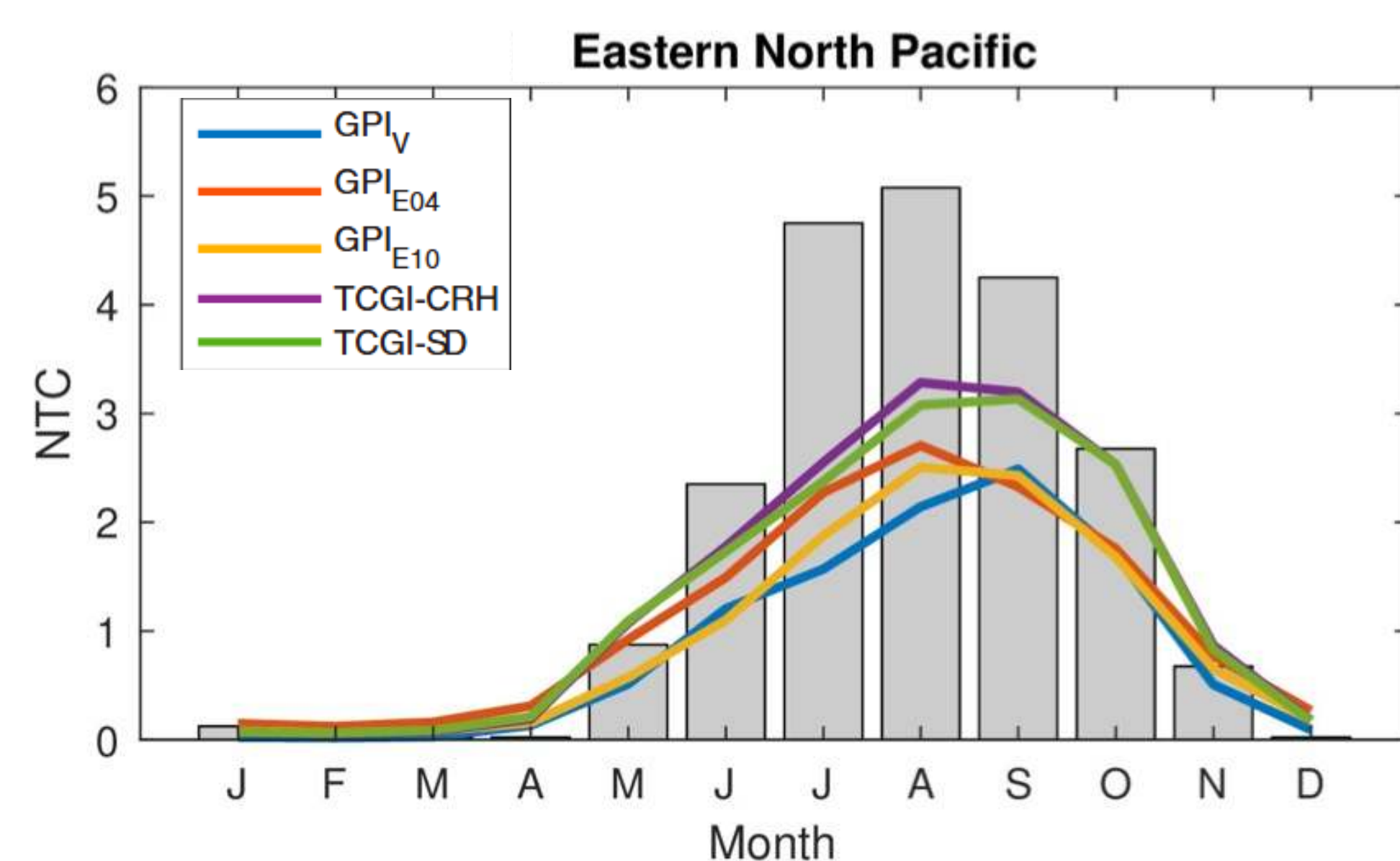


Fig1. Sketch (enhanced by AI) of Tropical Cyclone genesis.

- Recent Studies (Patricola et al., 2018; Danso et al., 2022) have shown that African Easterly Waves don't control the number of TCs in the Atlantic.
- The ITCZ in the Eastern and Central Pacific (ECP) is highly active during summer and fall, and it can undergo breakdown either by itself or by Tropical Easterly Waves (TEWs). (Wang & Magnusdottir, 2006).
- Large-scale environmental parameters, such as the Genesis Potential Index (GPI), are not effective in predicting storm hotspots in the ECP, regardless of the reanalysis dataset or the specific formulation of the GPI used (Sobel et al., 2021; Chavas et al., 2025).

Fig 2. Annual Cycle of TC genesis (gray bars) and 5 different normalized-GPIs (color lines, GPI_v in blue). Figure 11f from Chavas et al. (2025)

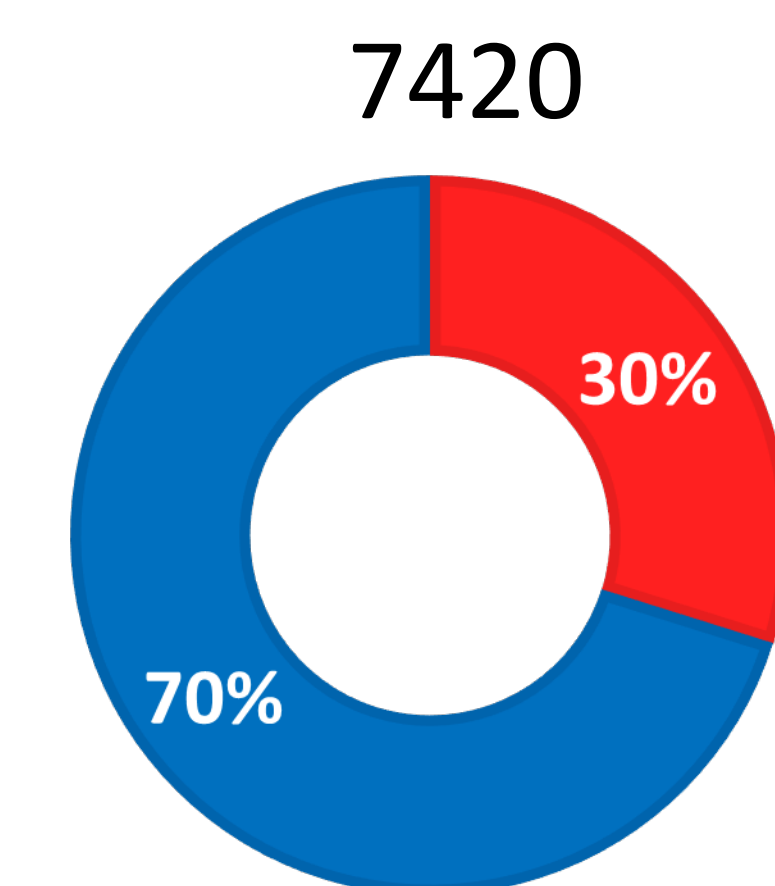
Research Questions

- Does seed structure vary significantly based on the disturbance that produces it? If so, is there a better seed structure?
- Does Non-TEWs contribute significantly to the number of TCs in the ECP?

Data and Methods

- For **seeds**, we use the database from Moon et al., 2025, that tracks seeds based on **geopotential at 850 hPa**.
- For **TEWs**, we use the database from Hollis et al., 2024, that track seeds based on **curvature vorticity at 700 hPa**.
- A **seed is associated to a TEW** if it is within a Greater Circle of **2000km at time of genesis**.

Seeds in the ECP



Developers in the ECP

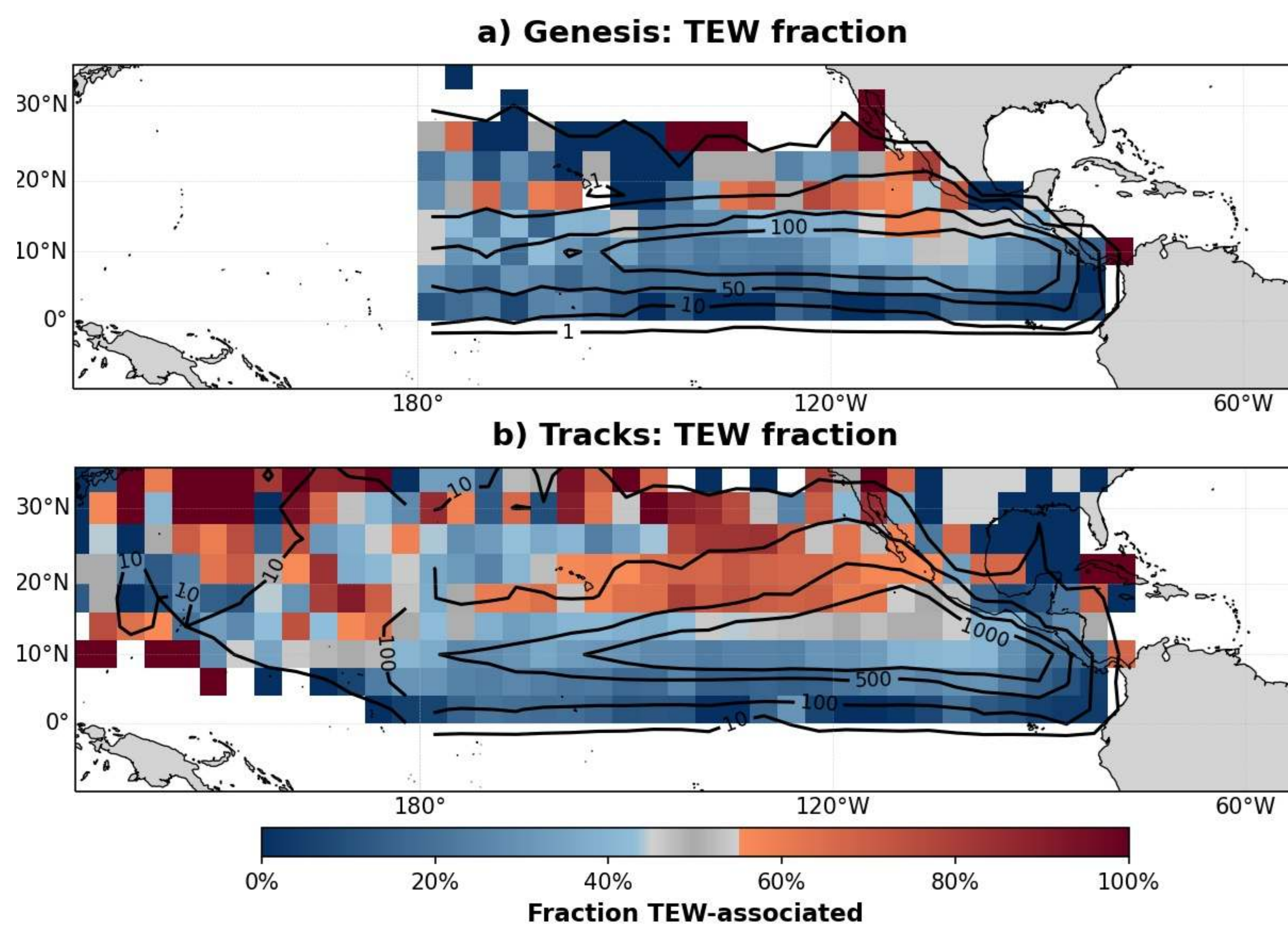
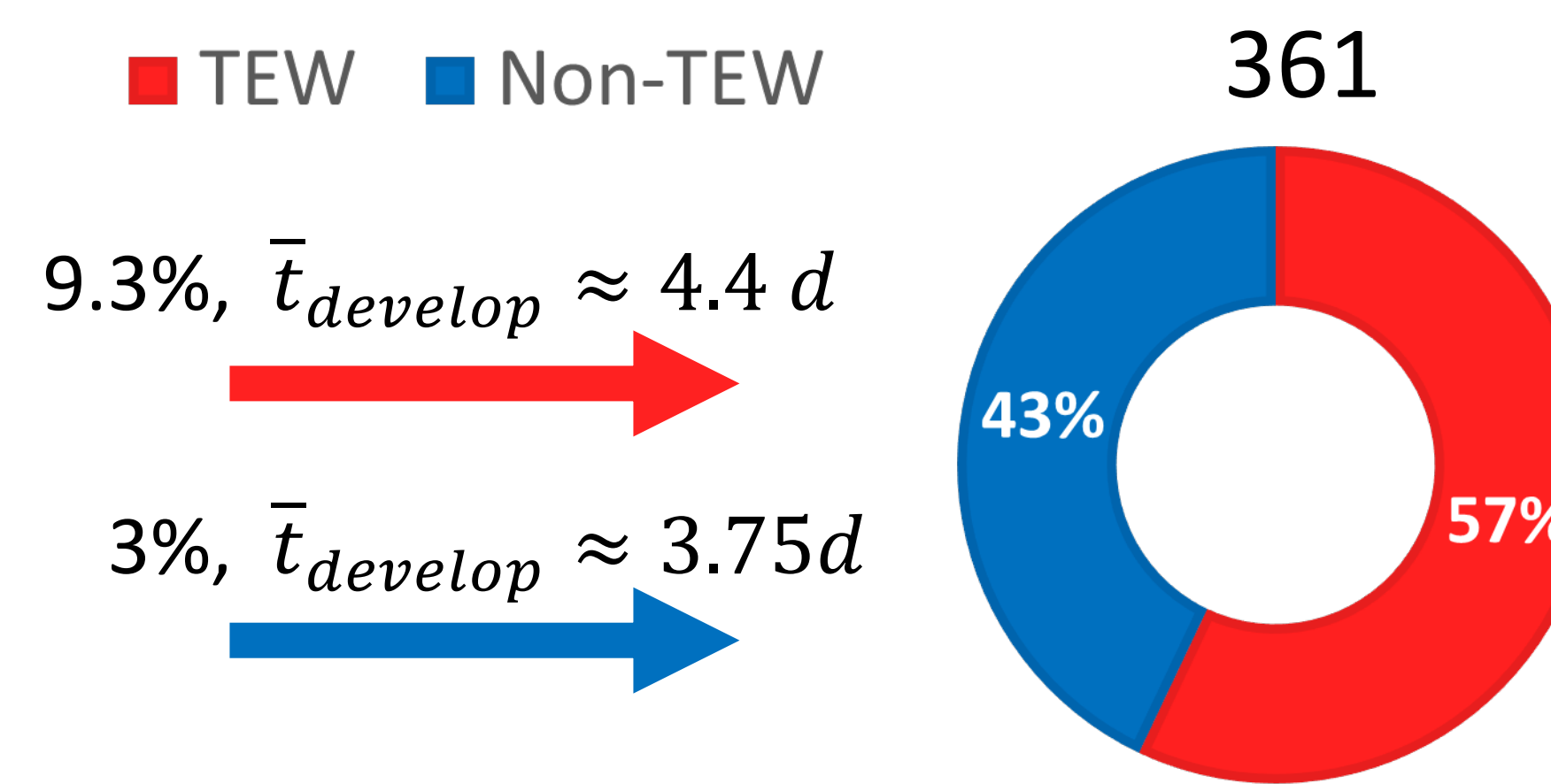
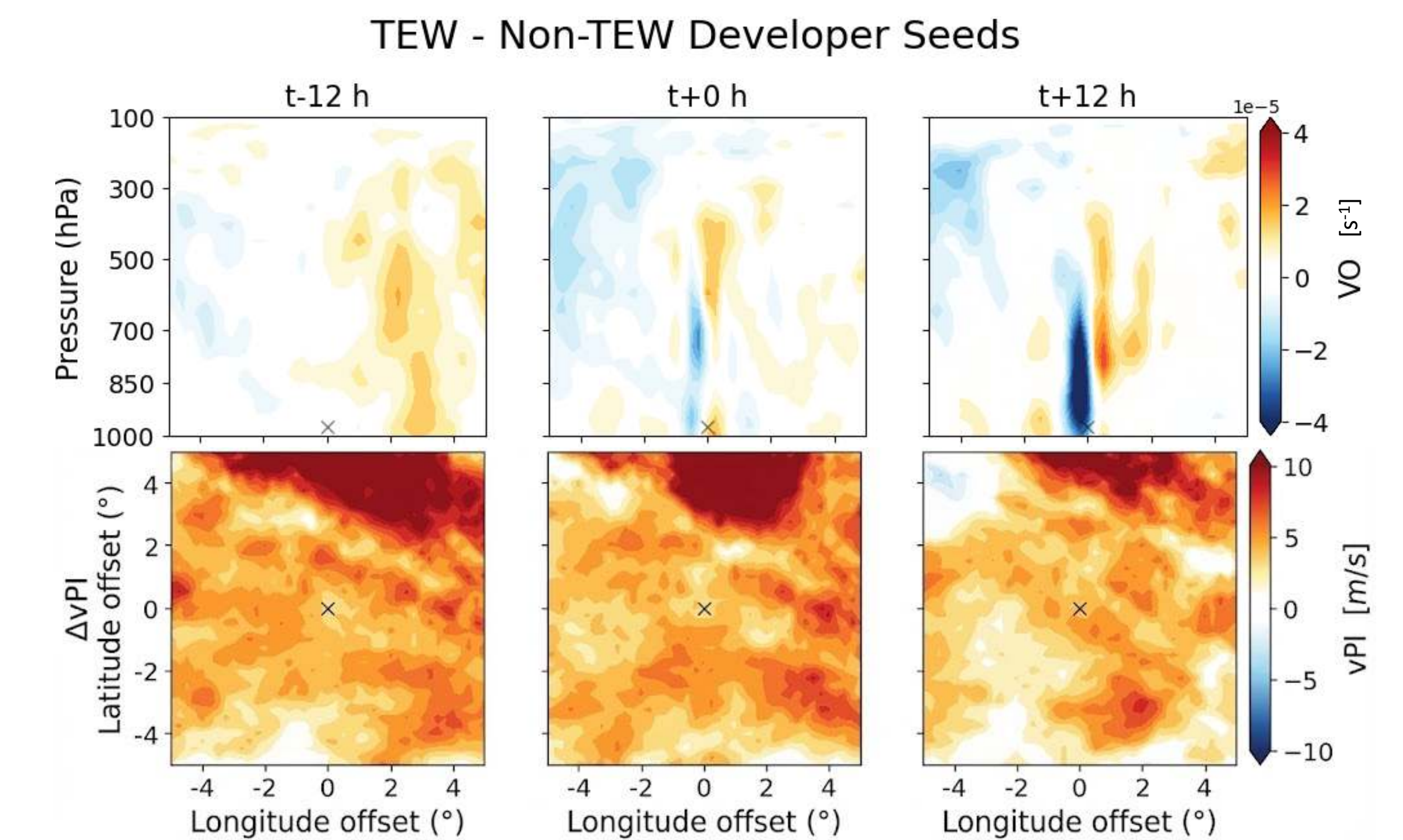


Fig 3. Percentage of seeds associated to TEWs in the ECP. Red regions are regions dominated by TEW seeds, while blue regions are dominated by non-TEW seeds. Contours are respective densities

Preliminary results

Non-TEW are more bottom heavy, but tend to live in a worse thermodynamic environment

Fig 4. Differences between TEW and non-TEW seeds in the ECP. First row show the delta of relative vorticity anomalies(s⁻¹), and the second row for **ventilated Potential Intensity** (m/s).

Developers Stand Out Early: Stronger Convection and Rain

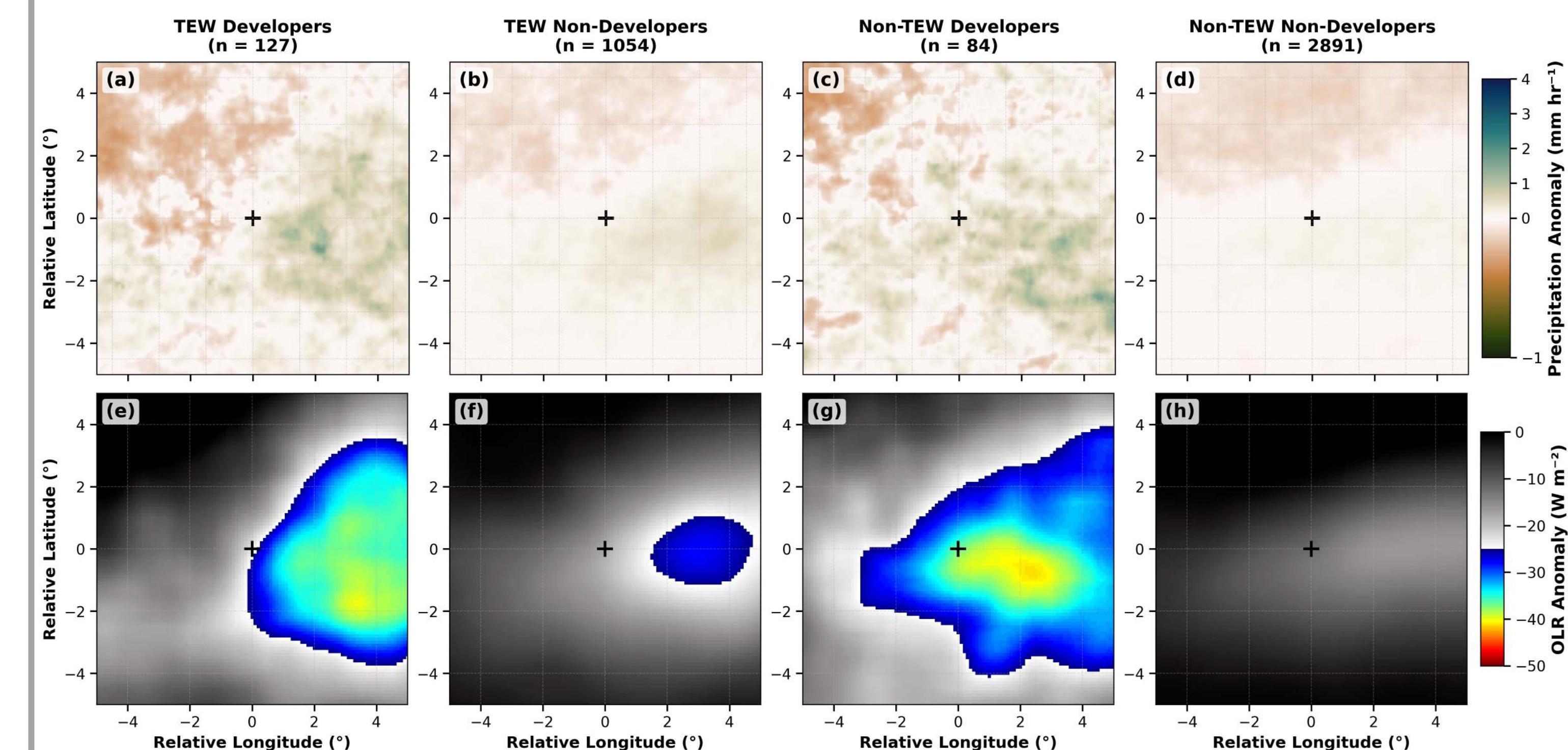


Fig 5. Composites of observational anomalies for TEW and Non-TEW seeds, 18 hours before their first detection. (a-d) Show Precipitation Anomalies from IMERG and (e-h) OLR anomalies from CERES

TEWs produce stronger TCs due to a better environment

LMI Distribution for TCs forming from a TEW and Non-TEW Seeds

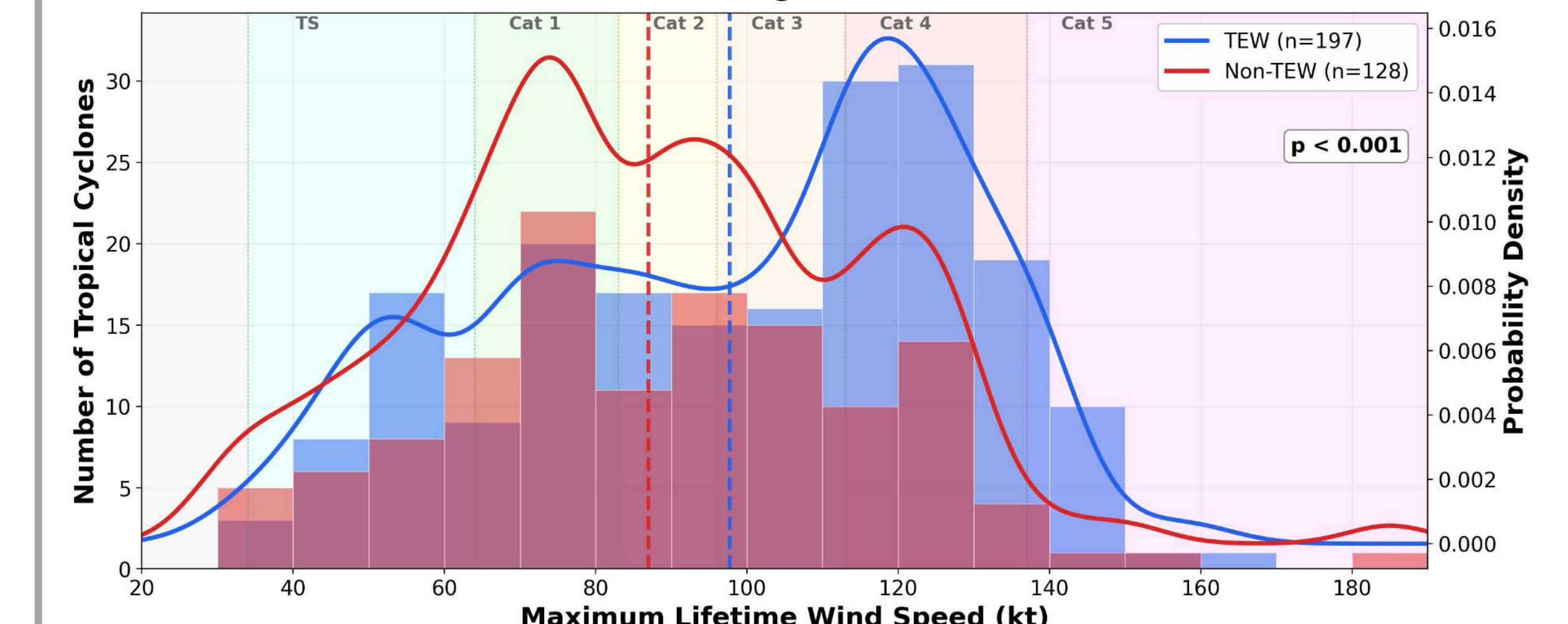


Fig 6. Distribution of Lifetime Maximum Intensity (LMI) for TEW and Non-TEW

